CE 222 – *Computer Organization and Assembly Language (COAL)*
Lecture 38

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Lecture Outline

- Announcements (Assignment and Quiz 5)
- Outline of Previous Lecture:
  - **Chapter 5 – Large and Fast: Exploiting Memory Hierarchy**
    - 5.2 Memory Technologies
    - 5.3 The Basics of Caches

- Outline of Today's Lecture:
  - **Chapter 5 – Large and Fast: Exploiting Memory Hierarchy**
    - 5.3 The Basics of Caches

Chapter 5

Large and Fast: Exploiting Memory Hierarchy

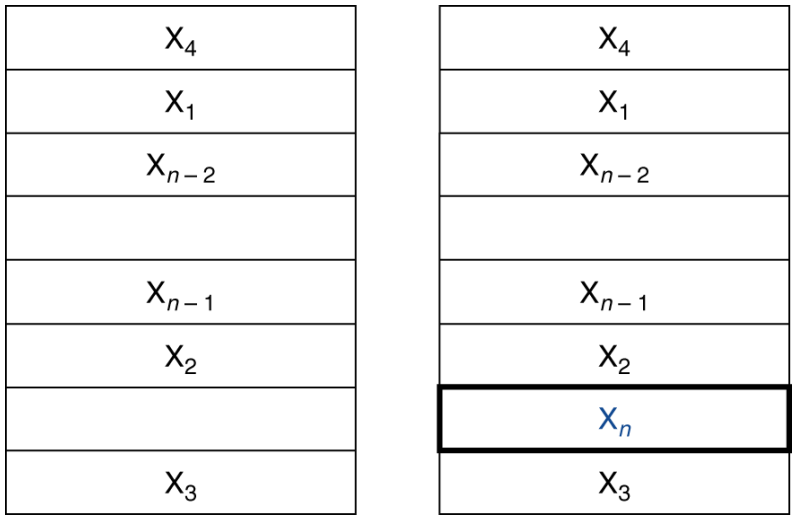
Chapter Outline

Chapter 5 – Large and Fast: Exploiting Memory Hierarchy

- 5.1 Introduction
- 5.2 Memory Technologies
- 5.3 The Basics of Caches
- 5.4 Measuring and Improving Cache Performance
- 5.5 Dependable Memory Hierarchy
- 5.6 Virtual Machines
- 5.7 Virtual Memory
- 5.8 A Common Framework for Memory Hierarchy
- 5.6 onwards (self-reading)

Cache Memory

- Cache memory
  - The level of the memory hierarchy closest to the CPU
- Assume if the cache contains recent data accesses X_1, …, X_{n-1}, and the processor requests X_n

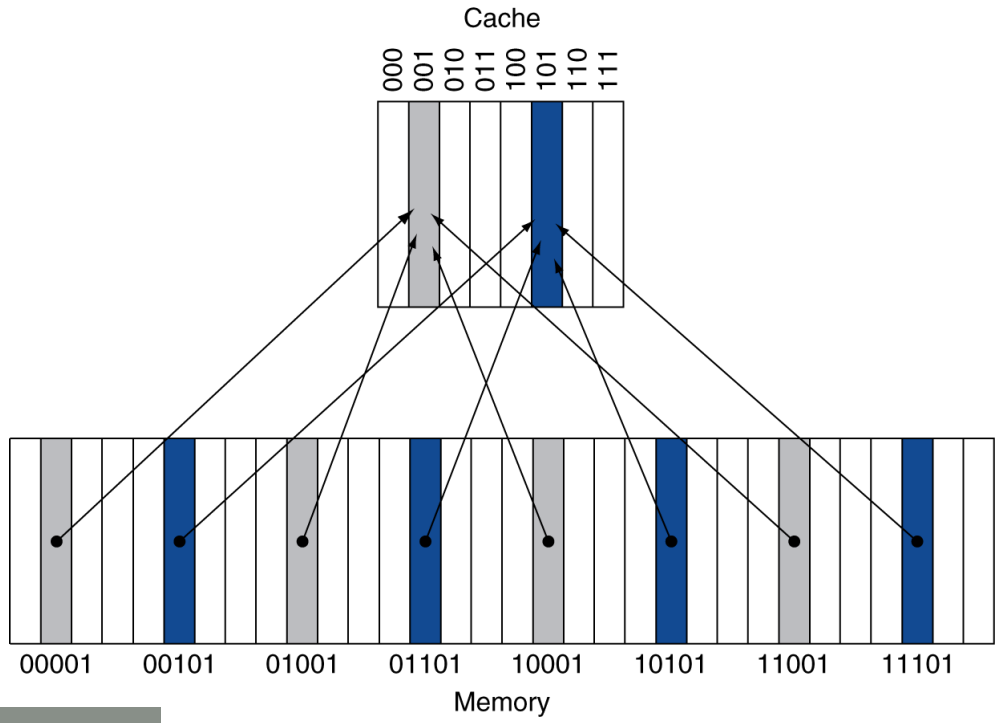


a. Before the reference to X_n b. After the reference to X_n

- How do we know if the data is present?
- Where do we look?

Direct Mapped Cache

- A good strategy would be to fix the cache location of data based on its memory address
- Direct mapped: only one choice
  - (Block address) modulo (#Blocks in cache)



- If #Blocks is a power of 2, one can use low-order log2(#Blocks) address bits

Tags and Valid Bits

- How do we know which particular block is stored in a cache location?
  - Store the high-order bits from block-address as well
  - Called the tag
- What if there is no valid data in a location?
  - Valid bit: 1 = present, 0 = not present
  - Initially 0

Cache Example

- Consider an 8-blocks, 1 word/block, direct mapped cache
- Below is a sequence of nine memory references to the initially empty cache

Decimal address of reference	Binary address of reference
22	10110_{two}
26	11010_{two}
22	10110_{two}
26	11010_{two}
16	10000_{two}
3	00011_{two}
16	10000_{two}
18	10010_{two}
16	10000_{two}

Cache Example

- Initial state

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	N		
111	N		

Cache Example

Word addr	Binary addr	Hit/miss	Cache block
22	10 110	Miss	110

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	Y	10	Mem[10110]
111	N		

Cache Example

Word addr	Binary addr	Hit/miss	Cache block
26	11 010	Miss	010

Index	V	Tag	Data
000	N		
001	N		
010	**Y**	**11**	**Mem[11010]**
011	N		
100	N		
101	N		
110	Y	10	Mem[10110]
111	N		

Cache Example

Word addr	Binary addr	Hit/miss	Cache block
22	10 110	Hit	110
26	11 010	Hit	010

Index	V	Tag	Data
000	N		
001	N		
010	Y	11	Mem[11010]
011	N		
100	N		
101	N		
110	Y	10	Mem[10110]
111	N		

Cache Example

Word addr	Binary addr	Hit/miss	Cache block
16	10 000	Miss	000
3	00 011	Miss	011
16	10 000	Hit	000

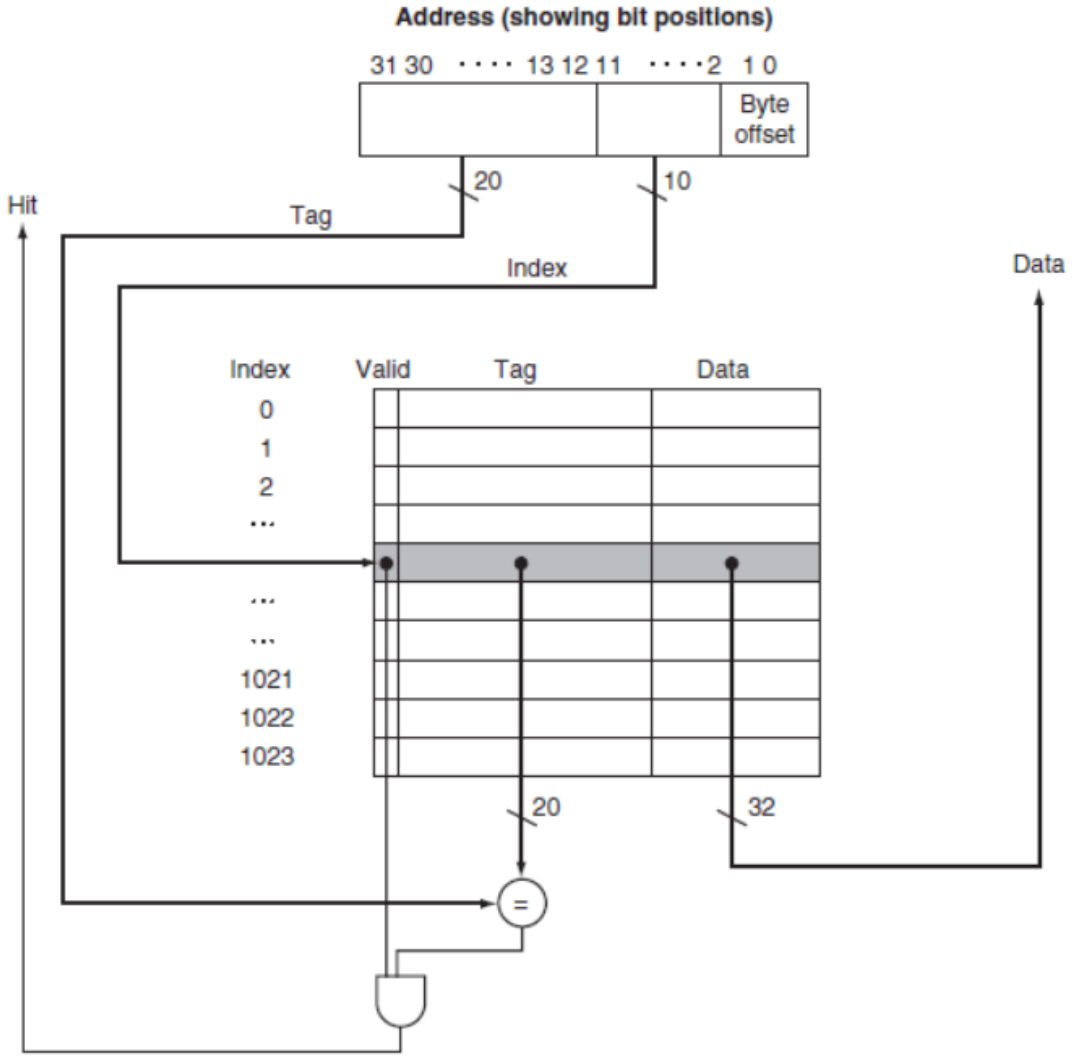
Index	V	Tag	Data
000	Y	10	Mem[10000]
001	N		
010	Y	11	Mem[11010]
011	Y	00	Mem[00011]
100	N		
101	N		
110	Y	10	Mem[10110]
111	N		

Cache Example

Word addr	Binary addr	Hit/miss	Cache block
18	10 010	Miss	010

Index	V	Tag	Data
000	Y	10	Mem[10000]
001	N		
010	**Y**	**10**	**Mem[10010]**
011	Y	00	Mem[00011]
100	N		
101	N		
110	Y	10	Mem[10110]
111	N		

Address Subdivision



Example: Total bits in Cache

- Consider:
  - 32-bit address size
  - Direct-mapped cache
  - Cache size = 2^n blocks
  - Block size = m words
  - Valid field size = 1 bit

- The naming convention is to exclude the size of the tag and valid field and to count only the size of the data.

- Example: How many total bits are required for a direct-mapped cache with 16 KiB of data and four-word blocks, assuming a 64-bit address?

Example: Mapping Address to Multiword Cache

- Example: Mapping Address to Multiword Cache Block
- 64 blocks, 16 bytes/block
  - To what block number does *address* 1200 map?

Block Size Considerations

- Larger blocks should reduce miss rate due to better exploitation of spatial locality

- But in a fixed-sized cache
  - Larger blocks ⇒ fewer blocks in the cache ⇒ More competition ⇒ increased miss rate
  - Larger blocks ⇒ pollution

- Larger miss penalty
  - Can override benefit of reduced miss rate
  - Early restart and critical-word-first can help

Handling Cache Misses

- On cache hit:
  - Pipelined CPU proceeds normally
- On cache miss:
  1. Stall the CPU pipeline instead of an exception/interrupt
     - Some advanced out-of-order processors continue with other instructions
  2. Fetch block from next level of hierarchy
     - What memory address should one use for Instruction cache miss
  3. Re-perform the task
     - If Instruction cache miss
       - Restart instruction fetch
     - If Data cache miss
       - Complete data access

Handling Writes

- On *data-write hit*:
  - Just update the block in cache
  - Can result in the inconsistency problem!
  - **Solution 1: Write through**: also update memory
- But writes take longer
  - e.g., if base CPI = 1, 10% of instructions are stores, write to memory takes 100 cycles
    - Effective CPI?
  - **Solution 2: Write buffer**
    - Holds data waiting to be written to memory
    - CPU continues immediately
- Pros/Cons ?

Handling Writes

- **Solution 3: Write-Back**:
  - On data-write hit, just update the block in cache
  - Declare that block ‘dirty’ and keep track of whether or not a block is dirty
  - When a dirty block is replaced
    - Write it back to memory
    - Can use a write buffer to allow replacing block to be read first
  - Pros/Cons ?